Global Renewable Energy in 2025: How Far Have We Come?

For policy analysts and energy sector strategists

A data-forward review of global renewable progress and gaps across primary energy, electricity, technology, and regions.

2024–2025 snapshot from Our World in Data

Source: Our World in Data (Ember 2026; Energy Institute 2025)

Fossil Fuels Drive ~75% of Global GHG Emissions and ≥5 Million Prematur

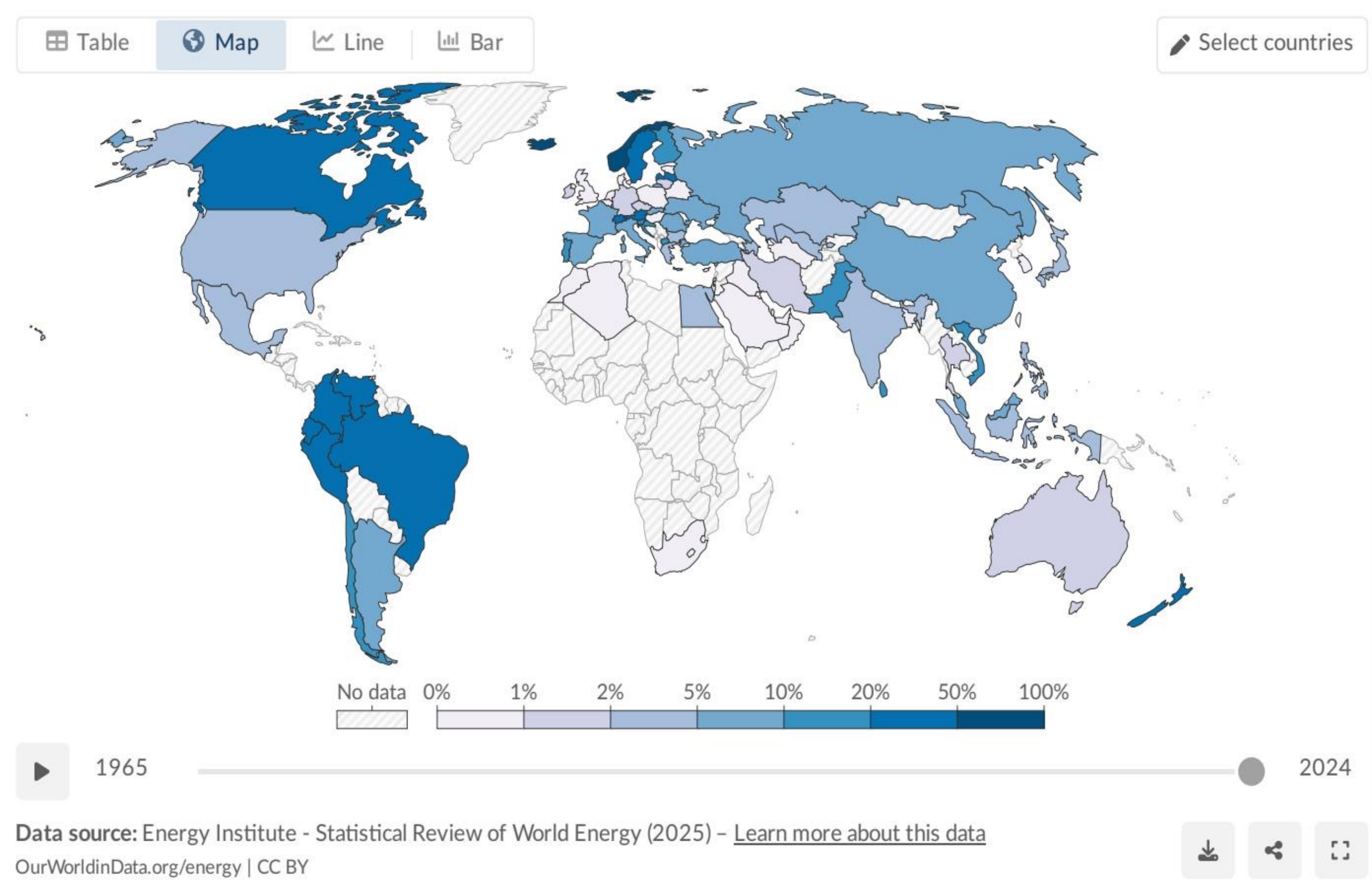
• Since the Industrial Revolution, energy systems became dominated by coal, oil, and gas.
• Burning fossil fuels accounts for roughly three-quarters of global greenhouse gas emissions.
• Combustion-related air pollution causes at least 5 million premature deaths each year.
• Rapid scaling of low-carbon energy—renewables and nuclear—is central to reducing climate and health harms.

Why this matters now

Renewables are no longer niche—they are a core pillar of energy security, affordability, and decarbonization.

Source: Our World in Data (Ember 2026; Energy Institute 2025)

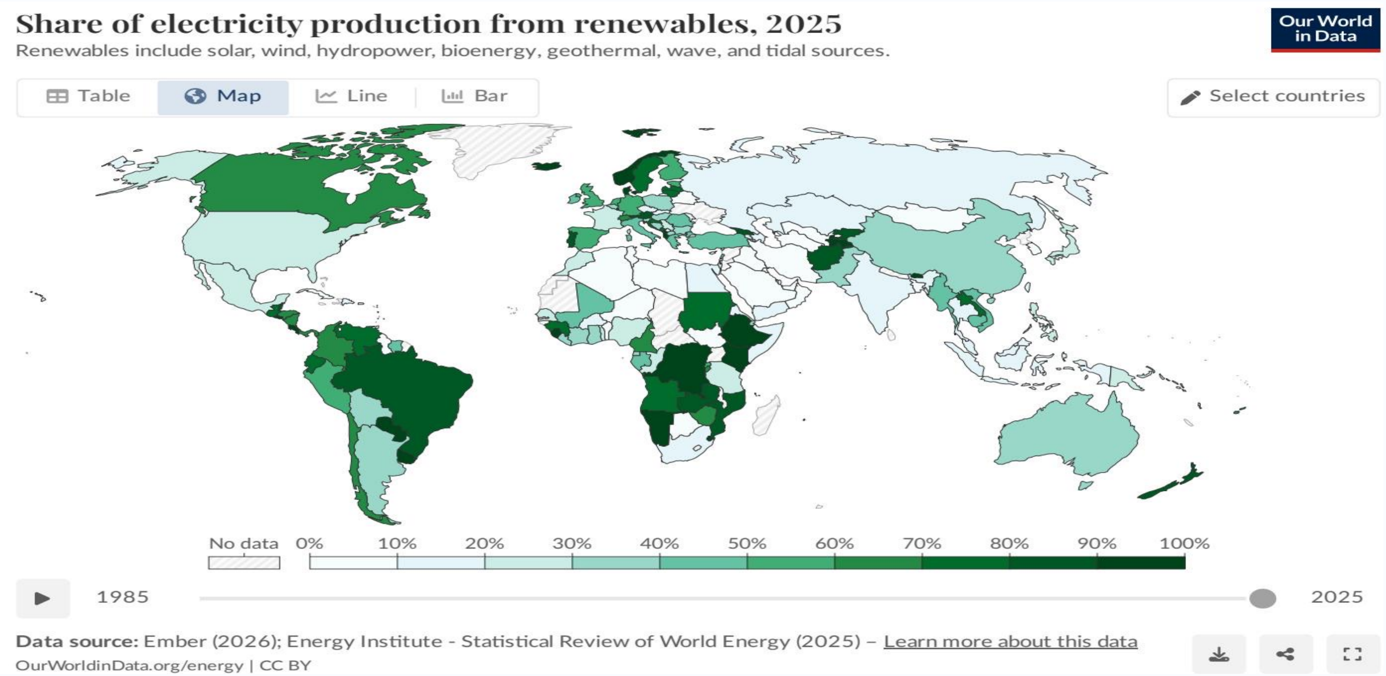
Renewables Provide ~14% of Global Primary Energy (2024), Highlighting S



- Primary energy includes electricity, transport, and heating.
- Global renewable share is around one-seventh (~14%).
- Transport and heating remain harder to decarbonize, keeping total share below electricity-only progress.
- High-share examples include Norway, Brazil, and New Zealand; some Middle Eastern producers remain below 5%.

Source: Our World in Data (Ember 2026; Energy Institute 2025)

Electricity Is Near One-Third Renewable Globally (~30% in 2025), But Cou

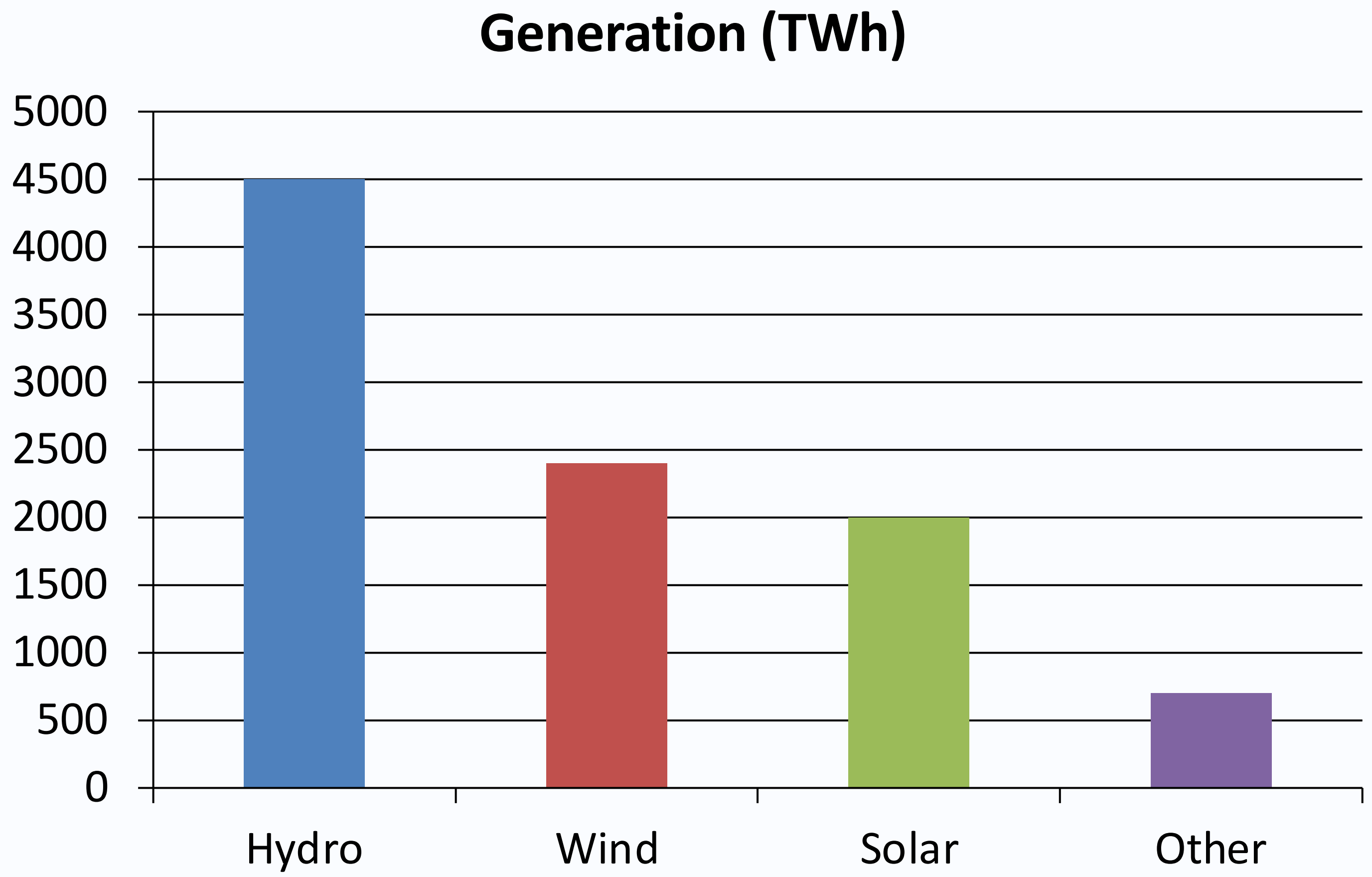


Country/Region	Renewable electricity share
Brazil	~85%
Canada	~65%
Norway	~98%
China	~35%
India	~25%
Saudi Arabia	<2%

Many African and Middle Eastern countries remain below 10%, while hydropower-rich systems and early clean-power investors lead.

Source: Our World in Data (Ember 2026; Energy Institute 2025)

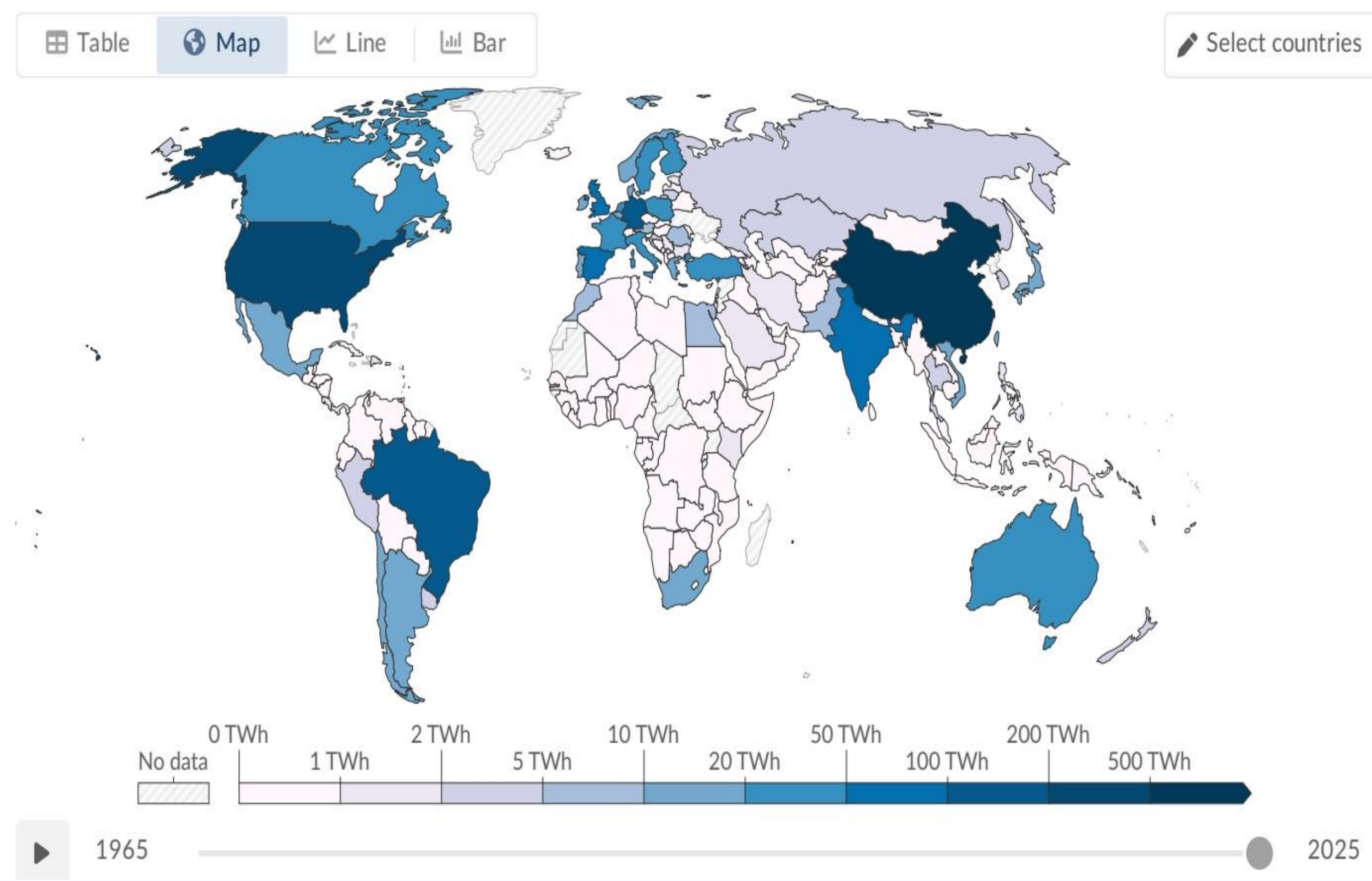
Renewable Electricity Totals ~9,600 TWh: Hydro Leads, Wind and Solar Cl



Technology	TWh	Share
Hydropower	~4,500	~47%
Wind	~2,400	~25%
Solar	~2,000	~21%
Other	~700	~7%

Source: Our World in Data (Ember 2026; Energy Institute 2025)

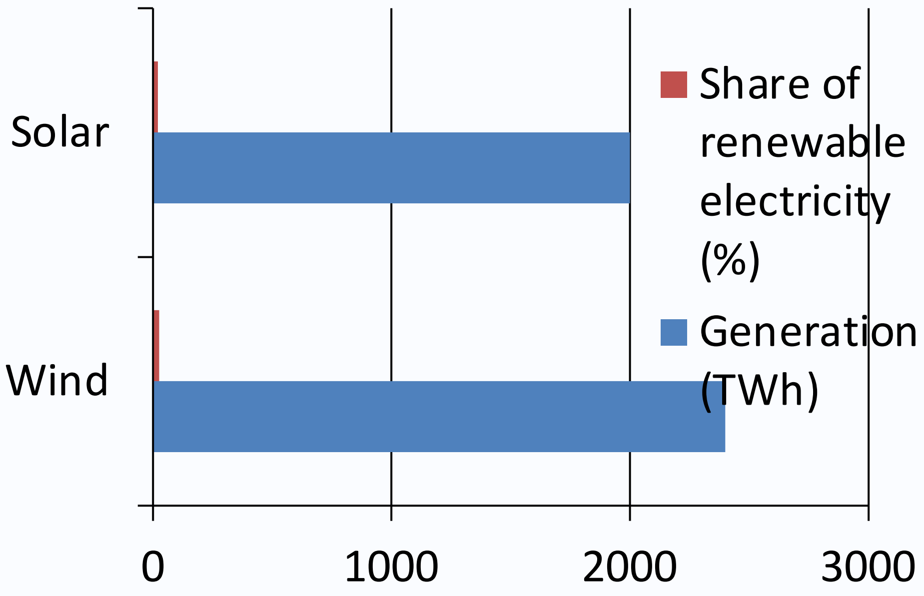
Hydropower Still Supplies Roughly Half of Renewable Electricity (~4,500 T



- Largest renewable source globally.
- Long-standing role in system balancing and firm generation.
- Top producers include China, Brazil, and Canada.
- Future growth is more geographically constrained than wind and solar.

Source: Our World in Data (Ember 2026; Energy Institute 2025)

Wind and Solar Are the Fastest-Growing Renewable Sources (Wind ~2,400



Together, wind and solar now generate nearly as much renewable electri

Source: Our World in Data (Ember 2026; Energy Institute 2025)

Regional Disparities Persist: Leaders Above 60–90%, Laggards Often Belov

• Scandinavia, Brazil, and Canada benefit from abundant hydropower resources.
• Countries such as Denmark and Germany accelerated wind and solar via sustained policy support.
• Much of Sub-Saharan Africa remains constrained by limited infrastructure and finance.
• Many Middle Eastern systems are still overwhelmingly fossil-fuel dependent.

Source: Our World in Data (Ember 2026; Energy Institute 2025)

Key Takeaways: Renewables Are Advancing Quickly, but Overall Transition

- ~14% of global primary energy and ~30% of global electricity now come from renewables.
- Hydropower remains the largest contributor (~47%), but wind and solar are catching up rapidly.
- Wind and solar together now generate nearly as much renewable electricity as hydropower.
- Regional disparities remain stark, with many countries still below 10% renewable electricity.
- Electricity is leading; faster progress in transport and heating is essential for whole-system decarbonization.

Policy implication: sustaining momentum now requires faster grid build-out, investment mobilization, and sector coupling beyond power.

Source: Our World in Data (Ember 2026; Energy Institute 2025)